Transfer Function: S Domain to Z Domain

1 S Domain

$$H_{model}(s) = \frac{1.1592 \times 10^7}{s^2 + 6.6172 \times 10^3 \cdot s + 1.1592 \times 10^7} \tag{1}$$

$$H_{lpf}(s) = \frac{6.2832 \times 10^4}{s + 6.2832 \times 10^4} \tag{2}$$

$$\begin{aligned} H(s) &= H_{model}(s) \cdot H_{lpf}(s) \\ &= \frac{7.2835 \times 10^{11}}{s^3 + 6.9449 \times 10^4 \cdot s^2 + 4.2736 \times 10^8 \cdot s + 7.2835 \times 10^{11}} \end{aligned} \tag{3}$$

2 Z Domain

$$H(z^{-1}) = z^{-1} \cdot \frac{1.0273 \times 10^{-4} \times (1 + 3.1629z^{-1})(1 + 0.2235z^{-1})}{(1 - 1.9348z^{-1} + 0.9360z^{-2})(1 - 0.5335z^{-1})} \tag{4}$$